PHY4605 Computational Methods in Physics
Learning Note | Week 10

Random Sampling and Monte Carlo: From One Random Path to Reliable Evidence

Use repeated random trials to describe physical variability, test reproducibility, and see how sample size affects a Monte Carlo estimate.

Physical question. A small particle receives random left-or-right kicks. One path is unpredictable, but if we simulate many particles under the same rule, what stable pattern can we extract from their final positions?

Core learning outcomes

1. explain what one random trial represents physically and reproduce a supplied random-sampling scaffold;
2. calculate and interpret the mean and spread of repeated outcomes; and
3. compare several sample sizes, use a fixed random seed when required, and validate the result against the physical model.

1. A random model still needs a physical rule

The lecture model is a one-dimensional symmetric random walk. A particle starts at

$$x_0 = 0\,\text{mm}.$$

Every second it moves exactly one step of length

$$a = 1\,\text{mm},$$

either to the right or to the left with equal probability. After $N = 100$ steps, the particle has one final position. We do not know that position in advance because the sequence of directions is random.

The randomness is bounded by a precise model: every step has the same magnitude, the two directions are equally likely, and successive steps are treated as independent. If any of those assumptions changes, the expected distribution of final positions also changes.

Prediction checkpoint. The model has no preferred direction. Across many independent particles, the average final position should therefore be near $0\,\text{mm}$. Individual particles should still finish at many different positions.

2. What exactly is one Monte Carlo trial?

For this model, **one trial is one complete 100-step path**. One trial contains many random draws, but it produces one main repeated outcome: the final position after 100 steps.

Repeating the Monte Carlo experiment means simulating many independent 100-step paths under the same physical assumptions. The number of repeated paths is the Monte Carlo sample size M.

Keep these two counts separate:

Quantity	Week 10 value	Meaning
N	100 steps/trial	Physical duration of one random path
a	1 mm/step	Physical step length
M	100, 1000, or 10000 trials	Number of independent paths used for the estimate

Increasing N changes the physical walk itself. Increasing M gives more repeated evidence about the same 100-step model.

3. Turn random draws into physical steps

The computational rule is simple:

1. choose N, a, M, and a random seed when an exact rerun is required;
2. generate one random number for every step of every trial;
3. map values below 0.5 to $-a$ and values at least 0.5 to $+a$;
4. cumulatively sum the steps to obtain trajectories;
5. extract the final position from each trial;
6. calculate the mean and spread of the final positions;
7. compare summaries for several values of M; and
8. validate the numerical evidence against symmetry and the known step rule.

This is the same course reasoning chain used in earlier weeks: physical model $\to$ algorithm $\to$ code $\to$ numerical evidence $\to$ validation $\to$ interpretation.

4. MATLAB random-sampling scaffold

Base MATLAB provides the random numbers. The Week 10 learning target is reading and modifying the scaffold, not deriving a random-number generator.

```
n_steps = 100;
step_length_mm = 1;
maximum_trials = 10000;

rng(4605,'twister')
u = rand(n_steps,maximum_trials);
direction = 2*(u >= 0.5)-1;
step_mm = step_length_mm*direction;

trajectory_mm = cumsum(step_mm,1);
final_position_mm = trajectory_mm(end,:);
```

The expression `u >= 0.5` produces logical values. Multiplying by 2 and subtracting 1 converts them to -1 or $+1$. Multiplying by `step_length_mm` attaches the physical step scale.

> **Do not confuse the seed with a physical parameter.** The seed chooses a reproducible random-number sequence. It does not represent time, position, temperature, or a property of the particle. Changing the seed should change finite-sample details while preserving the statistical behaviour of a correct model.

5. One trajectory is one possible history

The first simulated path below is produced with the locked seed. Another valid seed can produce a very different-looking path.

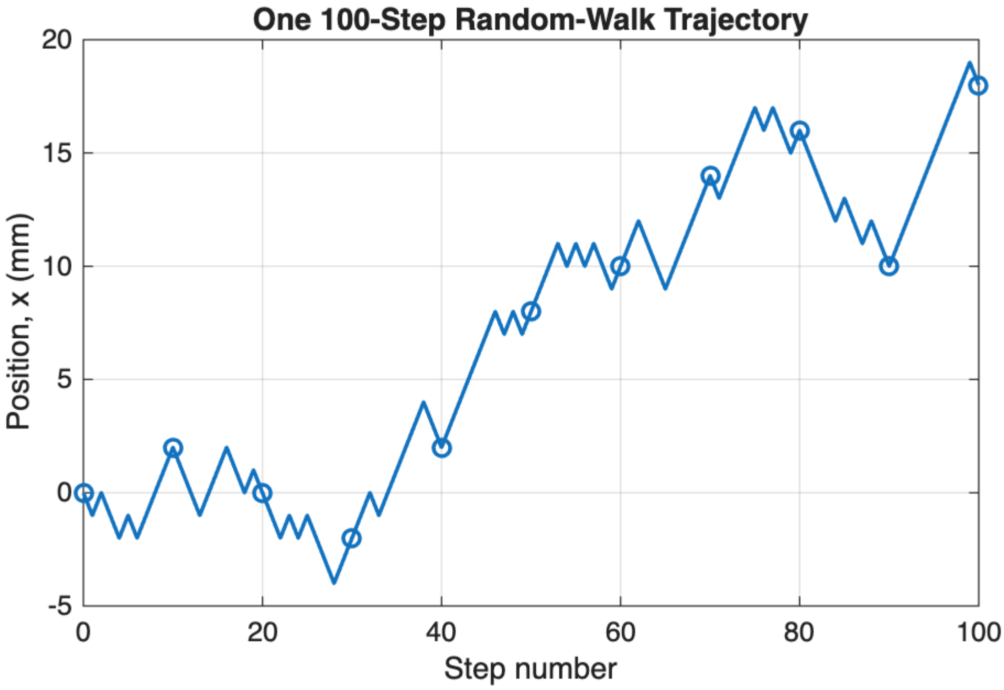

Figure 1: One 100-step random-walk trajectory. The path is physically possible under the model, but it is not the unique or "correct" path.

A trajectory is useful for connecting the code to motion, but one trajectory cannot tell us the long-run mean or spread. Those require repeated trials.

6. Repeated trials produce a distribution

With $M = 10000$ independent paths, the locked simulation gives a distribution of final positions.

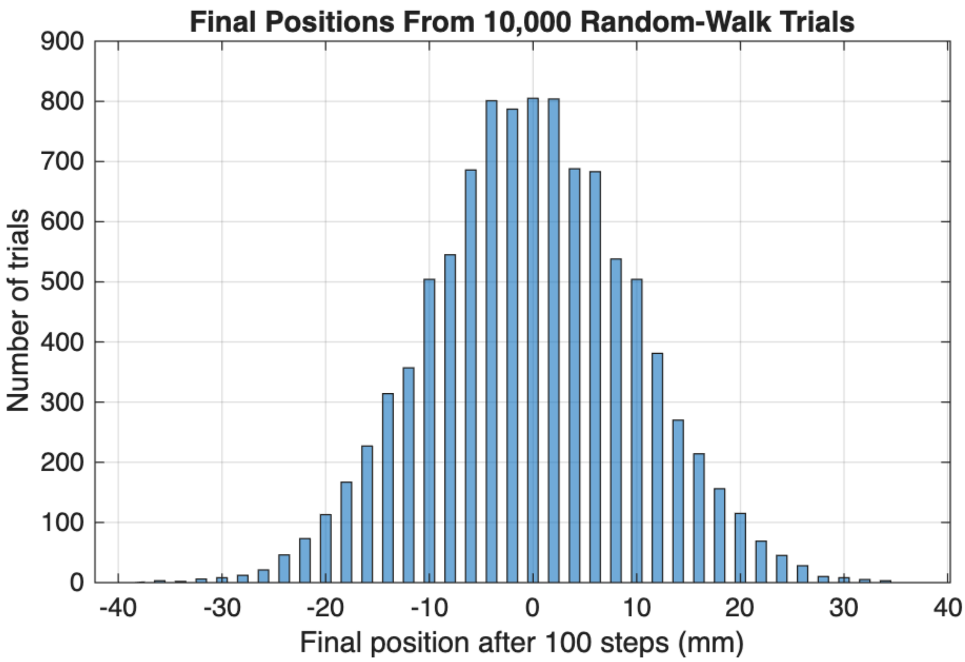

Figure 2: Final positions after 100 steps for 10000 repeated trials. The distribution is centred near zero but individual final positions vary widely.

For these 10000 trials,

$$\overline{x}_{\text{final}} = -0.128\,\text{mm}, \qquad s_{\text{final}} = 9.9423\,\text{mm}.$$

The **mean** describes the centre of the repeated outcomes. The **spread** describes how widely individual outcomes vary around that centre. A mean near zero does not mean every particle finishes at zero.

Core interpretation check. If the final-position mean is close to 0 mm but the spread is about 10 mm, the two results are not contradictory. Symmetry controls the centre; accumulated random steps create the variability.

7. Sample size changes the estimate, not the physical walk

Use the first 100, 1000, and 10000 trials from the same reproducible simulation:

Repeated trials M	Mean final position (mm)	Spread (mm)
100	0.040	10.6320
1000	0.066	9.8122
10000	-0.128	9.9423

The mean does not move monotonically toward zero: 0.040, 0.066, then -0.128 mm. Random convergence can fluctuate. What matters is that all three values are small compared with the roughly 10 mm spread and are consistent with a centre near zero.

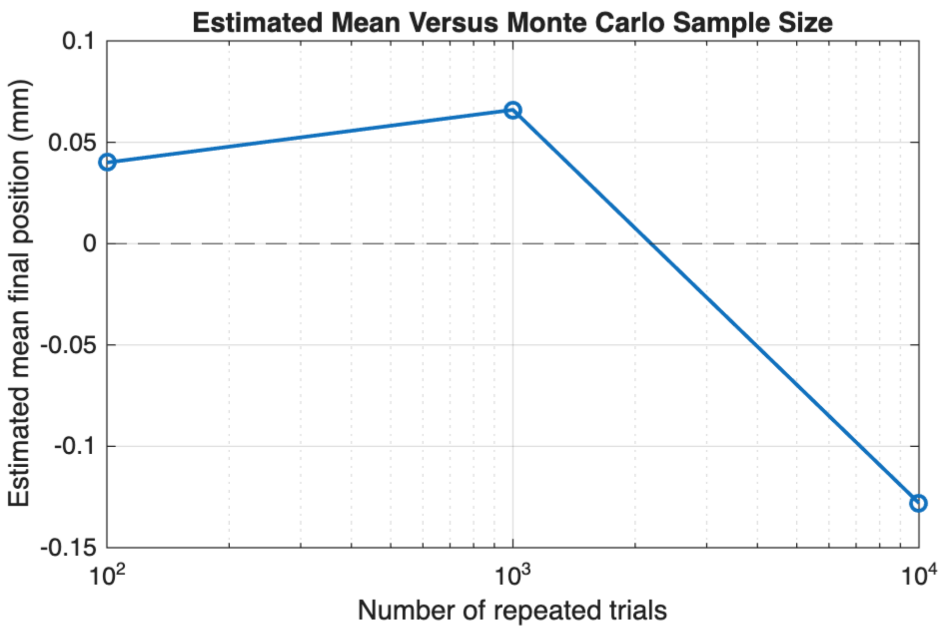

Figure 3: The finite-sample mean changes as more trials are included. More trials generally stabilise an estimate, but one random sequence need not approach the reference monotonically.

The spread stays near 10 mm for all three sample sizes because the underlying 100-step physical process has not changed. More Monte Carlo trials improve our evidence about the distribution; they do not make individual random walks less variable.

8. Validate with the model, not with appearance alone

Two Core checks follow directly from the physical rule.

Symmetry check. Equal left/right probabilities imply a long-run mean near 0 mm. The locked mean -0.128 mm is small compared with the width of the distribution.

Step-rule check. After exactly 100 steps of ± 1 mm, every final position must be an even integer number of millimetres. A result such as 7.3 mm would violate the model even if it appeared inside a plausible-looking histogram.

> **A reproducible wrong model is still wrong.** Resetting a seed can reproduce a coding defect perfectly. Always pair reproducibility with a physical or numerical validation check.

9. Worked reasoning chain

Step	Week 10 evidence
Physical model	symmetric 1D walk, 100 steps of ± 1 mm
Scale and units	position and step length in mm; trial count is dimensionless
Algorithm	random draw $\to$ direction $\to$ cumulative position $\to$ repeat
Code	supplied `rng`, `rand`, logical mapping, `cumsum`
Variability	distribution of final positions, mean, and spread
Validation	symmetry near zero plus even-integer final-position rule
Interpretation	more trials stabilise estimates without removing physical randomness

In an assessment, syntax may be supplied. You still need to define one trial, explain what repeated outcomes mean, keep physical units, distinguish mean from spread, and justify a validation check.

Working exposure: standard error and random-walk scaling

A supplied standard-error estimate for the mean is

$$\mathrm{SE}(\overline{x}) \approx \frac{s}{\sqrt{M}}.$$

For the locked simulation, the supplied values are about 1.063, 0.310, and 0.099 mm for $M = 100$, 1000, and 10000. Standard error describes precision of the *estimated mean*, not particle spread.

For the ideal symmetric walk, the supplied final-position scaling is

$$s_x \sim a\sqrt{N}.$$

With $a = 1$ mm and $N = 100$, this gives 10 mm; the simulated 9.9423 mm is close. Interpreting the scaling is Working exposure; deriving it is not required.

Optional stretch

Distribution derivations, statistical mechanics, variance reduction, MCMC diagnostics, and formal convergence theory are outside the Week 10 Core route.

Three takeaways

1. One trial needs a physical meaning; here it is one complete 100-step path.
2. Mean describes the centre; spread describes variability of individual outcomes.
3. A fixed seed supports reproducibility; sample-size and physical checks support trustworthiness.

> **Preparation for the practical.** Be ready to identify one random trial, trace a supplied random-sampling scaffold, interpret a histogram or summary table, compare sample sizes, use a fixed seed when needed, and validate the Monte Carlo result against independent physical evidence.